Q.1 $T: \mathbb{R}^4 \longrightarrow \mathbb{R}^3$

$T(e_1) = (1,1,1)$

$T(e_2) = (1,-1,1)$

$T(e_3) = (1,0,0)$

$T(e_4) = (1,0,1)$

Find $T(x,y,z,t)$?

A:- $\{e_1, e_2, e_3, e_4\}$ is LI.

Let $(x,y,z,t) = \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3 + \alpha_4 e_4$

$= (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$

$\Rightarrow \alpha_1 = x$

$\alpha_2 = y$

$\alpha_3 = z$

$\alpha_4 = t$

$\therefore (x,y,z,t) = xe_1 + ye_2 + ze_3 + te_4$

$T(x,y,z,t) = xT(e_1) + yT(e_2) + zT(e_3) + tT(e_4)$

$= x(1,1,1) + y(1,-1,1) + z(1,0,0) + t(1,0,1)$

$= (x+y+z+t, x-y+t, x+y+t)$

Thm - 4.5.1

Let $T: U \to V$ be a linear map and $\dim(U) = \dim(V) = n$. Then the following statements are equivalent.

(a) T is non-singular (an isomorphism)

(b) T is one-one

(c) T transforms linearly independent subsets of U into LI subsets of V.

(d) T transforms every basis for U into a basis for V.

(e) T is onto.

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